

Systematic Review

The Impact of ICT on Primary School Students' Natural Science Learning in Support of Diversity: A Meta-Analysis

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Abstract: In recent years, studies analyzing how information and communication technologies (ICTs) contribute to the development of students with special educational needs (SENs) have gained interest. The proliferation of studies in this field has driven the creation of numerous digital resources that can be applied in science teaching. Therefore, this study aims to conduct a systematic review of the literature on the usefulness of ICT in teaching Natural Sciences in Primary Education to address diversity. The methodology used for this systematic review and meta-analysis followed PRISMA guidelines, drawing data from the Scopus and Web of Science databases. A total of three documents were analyzed. The results indicated a favorable effect for the experimental groups ($I^2 = 84$; $p = 0.002$). These findings show that the use of ICT enhances participation and motivation among Primary School students with SENs in Natural Sciences. In conclusion, ICT positively influences learning in Natural Sciences by increasing motivation among Primary School students with SENs. This meta-analysis highlights the importance and positive impact of ICT on Natural Sciences learning in Primary Education, especially in support of student diversity. The reviewed evidence confirms that personalizing learning through adaptive systems, using methodologies based on individual learning styles, and employing innovative technologies significantly enhance academic performance, motivation, and student engagement. The effective use of ICTs for inclusion in primary school science relies on the design of instructional methods that link technology, emotions, and diversity.



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1. Introduction

The rapid emergence of Information and Communication Technologies (ICTs) has radically transformed our society and, in particular, the educational system (Camargo et al., 2025). This swift progress has led current regulations—such as Organic Law 2/2006 of May 3 on Education (LOE, 2006), as amended by Organic Law 3/2020 of December 29 (LOMLOE, 2020)—to promote a competency-based education in which the integration of ICT and the development of digital competence are fundamental pillars for preparing students to face the challenges of an increasingly digital environment. The LOMLOE, in line with the guidelines of international bodies such as the United Nations 2030 Agenda (UN, 2015), advocates for an inclusive and equitable approach to learning, promoting universal accessibility and high-quality education for all students.

In this context, inclusive education emerges as a fundamental paradigm, viewing diversity not as a difficulty, but as a value (Ratten, 2025; Uthus & Qvortrup, 2024). Following the Universal Design for Learning (UDL) approach, the goal is to eliminate physical,

emotional, and sensory barriers, ensuring access to the curriculum through active methodologies and the strategic use of ICT (Othman et al., 2023). Diversity, beyond being conceived exclusively from the perspective of special education (Caballero, 2019), encompasses a wide range of personal and social differences that require flexible, innovative, and adapted educational responses.

The implementation of ICT in the classroom has been identified as one of the main drivers toward inclusive education (Pegalajar, 2017). Recent research shows that its use not only serves as a tool for support and motivation but also enables the design of activities tailored to students' individual interests and abilities, thereby increasing their autonomy and active participation in the learning process (Fernández-Batanero et al., 2018; Martínez, 2020). Its impact is especially critical for students with special educational needs (SENs), who face greater socio-educational vulnerability (Tah, 2020).

Historically, *The Warnock Report* (1978) already highlighted the necessity of providing special means of curriculum access for these students. Today, the LOMLOE redefines SENs as needs arising from disability or severe behavior, language, or communication disorders, reaffirming the need for specific and ongoing support. Thus, “innovating to include”—coined as “inclunovation” by Elizondo (2019)—becomes an imperative that demands a transformative journey of our educational systems, as Arnaiz-Sánchez and Escarbajal (2021) point out. However, inclusion cannot be understood solely in terms of material resources; it also requires the fostering prosocial behaviors and emotional competencies in increasingly heterogeneous classrooms (Garrote et al., 2020; Fu et al., 2021). Collaboration among all educational agents, the use of cooperative methodologies, and the design of innovative activities are essential in improving coexistence and ensuring the participation of every student (Meindl et al., 2020; Barreiro, 2021).

There is a clear need for personalized emotional and academic support, fostering motivation, emotional involvement, and positive feedback cycles between teachers and students (López-Belmonte et al., 2022). Within this framework, methodologies such as Universal Design for Learning (UDL), combined with tools like augmented reality (AR), have proven especially effective in stimulating multisensory engagement, divergent thinking, and inclusion (Pastor, 2018; Marín et al., 2022). Likewise, acknowledging multiple intelligences (Gardner, 2017) and implementing a flexible curriculum, supported by technological resources, promote the holistic development of all learners (Badilla-Quintana et al., 2020).

Moreover, teaching practices should draw on frameworks such as Bloom and Krathwohl's (1956) taxonomy and focus on executive functions, which are fundamental for achieving curricular objectives and nurturing critical and creative thinking (Gómez-Tabares, 2022). However, challenges remain. The experiences of students with neurodevelopmental disorders, such as autism spectrum disorder (ASD), are still poorly understood and demand specific strategies—such as self-instruction techniques, gamification, or game-based learning (Rangvid, 2018; Dell'Armo & Tassé, 2019). This highlights the need for more in-depth research to reveal their realities and optimize inclusive educational processes (Zakai-Mashiach, 2022).

The learning of Science in Primary Education constitutes a fundamental foundation for the development of critical thinking, scientific curiosity, and the understanding of the natural and social environment (Ospankulova et al., 2024). In addition to promoting essential cognitive skills, this area represents a key opportunity to foster the active participation of all students, regardless of their individual characteristics (Muñoz-Losa & Corbacho-Cuello, 2025). In this regard, the incorporation of inclusive approaches in the teaching of Natural Sciences is crucial to ensuring equitable access to meaningful learning experiences, which, in turn, strengthens values such as diversity, respect, and democratic coexistence (Saha et al., 2022). Likewise, this educational stage is especially valuable for

introducing and consolidating the foundations of the STEM approach (Science, Technology, Engineering, and Mathematics), due to the intrinsic motivation and cognitive openness that characterize children at this age (Ospankulova et al., 2024), thereby promoting meaningful learning (Romero & Quesada, 2014).

However, the implementation of inclusive teaching in this field is not without challenges. Various studies indicate that one of the main obstacles lies in the insufficient teacher training in inclusive methodologies, which limits teachers' ability to adapt content to the diversity present in the classroom (Kratz & Schaal, 2015; Mafugu et al., 2022). To address this challenge, a pedagogical renewal in science education is required through truly transformative teacher training (García-García et al., 2019). This situation highlights the need to strengthen teacher preparation as an essential condition to achieve truly inclusive and high-quality science education, through a didactic model that promotes the development of a curriculum based on neuroscience, intercultural competencies, principles of sustainability, reflective practice, experiential learning, continuous professional development, and the application of inclusive pedagogical strategies (de Barros et al., 2024).

ICT not only facilitates access to information but also enhances students' autonomy, connection with their environment, and overall well-being (Soriano-Sánchez & Jiménez-Vázquez, 2022). Its effective implementation in Primary Education Science classrooms to address diversity thus emerges as a key factor for achieving inclusive, high-quality education in line with the principles of the 2030 Agenda. Despite these advances, significant challenges persist, requiring targeted teacher training, institutional commitment, and an educational vision that embraces diversity as a source of enrichment for the entire system. Despite the growing recognition of the potential of ICT to foster more inclusive educational environments, significant gaps remain in the literature regarding their specific effectiveness in supporting the learning of students with SENs. While many studies broadly address the use of ICT in diverse educational contexts, few offer a systematic and quantitative analysis that allows for solid conclusions about its actual impact on this particular group. This meta-analytic review is therefore justified by the need to synthesize the available evidence, identify patterns, and highlight both the progress and current limitations in the use of technological tools to support the learning of SEN students in Primary Education.

The Present Study

Based on the foregoing—and given that, to our knowledge, no systematic review and meta-analysis has yet examined the impact of ICTs use on the learning of Primary Education students with SENs—the aim of the present study is to conduct a systematic review of the literature focused on the usefulness of technological resources in teaching Natural Sciences to Primary Education pupils with SENs. Drawing on prior empirical evidence, the following research hypotheses (H) are proposed: **H1**. In the area of Natural Sciences, interactive and adaptive digital technologies are primarily used to support students with SENs; **H2**. The implementation of ICT in the teaching–learning process for students with SENs significantly improves their participation, motivation, and understanding of content. Finally, the following research question was posed: What thematic approaches prevail in the scientific literature on the use of ICT to promote inclusion in Primary Education within the area of Natural Sciences, according to a keyword co-occurrence analysis?

2. Materials and Methods

2.1. Information Sources and Search Strategy

The present study is based on a systematic review of the scientific literature. The guidelines proposed by the PRISMA statement (Page et al., 2021) were followed (Figure 1).



Figure 1. Summary of the steps to be performed in the systematic review with meta-analysis (PRISMA Statement).

The search was carried out in March 2025, on the Web of Sciences (WoS) and Scopus databases of the Elsevier group. These are the most prestigious scientific databases, framing the famous JCR and SJR impact indices, respectively. For the search, we used the filter of papers published in English and Spanish, as well as open-access studies. The search formula used was as follows: *(Natural Science*) AND (special educational needs*) OR (SEN*) AND (instruction*) AND (information communication technology*)*.

This formula was applied to the “title”, “abstract”, and “keywords” fields and adapted to the syntax of each database to ensure compatibility. The results obtained from each of the databases were as follows: 1282 in Web of Science and 3 in Scopus. In addition, manual checks of the references of the studies included were carried out to identify potentially eligible articles that were not captured in the initial database searches.

2.2. Procedure for Selection and Data Collection

The selection was performed by a single primary reviewer, and 20% of the studies were independently validated by an expert to reduce bias, although a full double review was not performed. However, although PRISMA recommends the involvement of at least two independent reviewers for study selection, this review relied primarily on one reviewer with partial validation to balance thoroughness and feasibility.

In turn, rigorous strategies were implemented to minimize selection bias, including the use of clearly defined inclusion criteria, systematic application of search terms across multiple databases, and manual reference checks. Additionally, consultation and supervision from an expert in the field were sought to ensure accuracy and consistency in the process, in line with PRISMA guidelines, which recommend the involvement of at least two reviewers. The selection process was carried out in several phases:

- Initial screening by title and abstract: One reviewer evaluated all identified records.
- Full-text evaluation: The preselected articles were read in their entirety to apply the inclusion and exclusion criteria. In case of doubts or ambiguities, a second methodological expert on the subject was consulted to ensure consistency in the decisions made.
- Duplicate detection: The Mendeley reference management system was used to organize the studies and eliminate duplicates.

Subsequently, the following inclusion and exclusion criteria were defined. Regarding the inclusion criteria, these were as follows: (a) empirical studies; (b) research that includes treatment with control and experimental groups; (c) articles that include means and standard deviations for both the control and experimental groups; (d) studies addressing the use of ICT in the science classroom with students with SENs; (e) research published between 2010 and 2025; (f) studies published in English and Spanish.

The exclusion criteria were as follows: (a) studies with restricted access to publication; (b) duplicate studies; (c) conference proceedings, book chapters, or books; (d) theoretical studies, reviews, case studies, or cross-sectional studies; (e) studies in a language other than Spanish or English; (f) studies irrelevant to the research topic (from other educational

stages or about issues unrelated to ICT or science); (g) lack of sufficient information for meta-analysis.

First, the titles and abstracts of the identified records were reviewed. The preselected records were obtained and read in full text. However, in cases where there was any controversy, a comprehensive reading of the full text was conducted to apply the remaining established conceptual and methodological criteria. The information from the records that met the eligibility criteria was extracted into a database.

The search formula used resulted in a total of 1285 documents (1282 from Web of Science) and (3 from Scopus). Regarding the exclusion criteria, a total of 1283 documents were discarded based on the established exclusion criteria, as follows: 1 study was excluded due to restricted access, according to exclusion criterion (a); 1 study was excluded according to exclusion criterion (b); 323 studies were excluded according to exclusion criterion (c); 67 studies were excluded based on exclusion criterion (d); 4 studies were excluded according to exclusion criterion (e), which corresponded to 13 studies published in Chinese, 7 published in Russian, and 2 published in Portuguese; 858 studies were excluded based on exclusion criterion (f); 4 studies were excluded according to exclusion criterion (g); finally, 2 studies were excluded due to a lack of results for inclusion in the meta-analysis, according to exclusion criterion (g). The flow diagram shows the process followed and the screening of the scientific articles, leading to the formation of the final sample. Finally, 17 studies met the eligibility criteria and were included for review. The selection process is summarized in the PRISMA flow diagram (Figure 2).

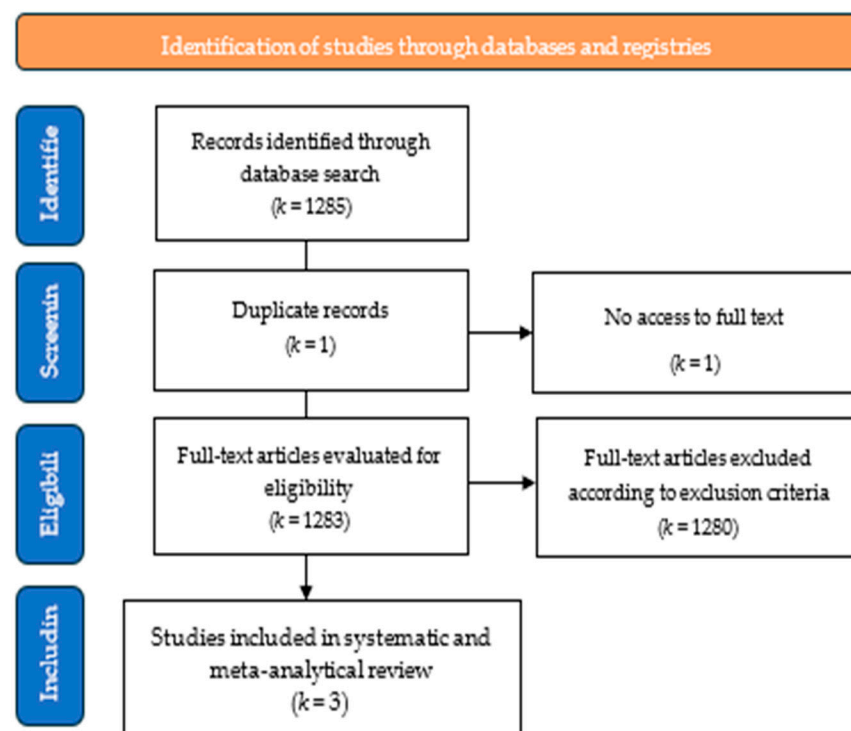


Figure 2. Filtering systematic review articles according to PRISMA flow (Page et al., 2021).

2.3. Data Extraction

For data extraction, an Excel sheet was designed that included information on the established inclusion criteria. These were coded following the process outlined below: (1) author(s); (2) year of publication; (3) study objective; (4) location; (5) educational stage; (6) study design; (7) participants in each group (experimental and control); (8) age range/mean age; and (9) treatment and intervention time (Table 1). With the help of the Excel sheet, the most relevant quantitative and qualitative data from each of the studies

were extracted. This process was carried out thoroughly, ensuring the highest reliability in data collection. This procedure was initially performed by one reviewer and subsequently randomly verified by a second independent reviewer to ensure data accuracy. The results were summarized in a summary table (Table 1), which presents the key characteristics of the included studies, as recommended by PRISMA.

Table 1. Summary of the studies included in the systematic review and meta-analysis.

Author(s) and Year of Publication	Study Objective	Place	Study Design	Participants in Each Group (<i>n</i>)		Age	ICT Employed
				Experimental	Control		
C. H. Chen et al. (2014)	Proposing a progressive, prompt-based contextual learning approach to improve the performance of students in a Primary School course	Taiwan	Experimental	30	30	12 years	Progressive, prompt-based mobile learning
M. H. M. Chen et al. (2019)	Explore the effects of scenario simulation games and electronic textbooks on the learning outcomes of Primary School students	Taiwan	Experimental	30	30	10–11 years	E-books and 3D and 4D video games
Lai et al. (2019)	Develop a science learning system using AR based on the contiguity principle of multimedia learning to promote students' science learning	Taiwan	Experimental	23	23	10–11 years	Traditional mobile augmented reality

2.4. Data Analysis: Meta-Analysis

Firstly, the option of Intervention Review was selected to test the effectiveness of the interventions, following a fixed-effects model, because it was assumed that the included studies share a common underlying effect and the differences observed among them are solely due to sampling error. The meta-analysis was conducted using the statistical program *Cochrane Review Manager* (RevMan), version 5.4 (Cochrane, London, UK), to check the heterogeneity of the studies, effect size, data quality, etc. (Sánchez-Meca & Ato, 1989), as it provides a high level of evidence on the effectiveness of the interventions. To analyze the data, the statistical method of standardized mean difference for random effects was used (*Fixed Effect*), since there were differences in the means of the included studies. The direction of the effect sizes is considered favorable if the result indicates an improvement in the set of interventions, with significance observed through the *p*-statistics, where the effectiveness of the set of interventions is considered significant if $p < 0.05$. Finally, regarding heterogeneity, it is considered high if $I^2 \geq 75\%$, moderate from 50 to 75%, and low when $I^2 \leq 25\%$ (Bölek et al., 2022; Higgins et al., 2003). Due to the limited number of studies, subgroup and sensitivity analyses were not feasible.

2.5. Protocol Registration

The protocol for this review was not registered with PROSPERO or any other repository due to the exploratory nature of the study and time constraints. However, this

omission is transparently reported, recognizing that protocol registration is a good practice recommended by PRISMA to ensure the traceability of the review process.

2.6. Risk of Bias Rating and Methodological Quality of Included Studies

The risk of bias was assessed by examining the distribution of points in the funnel plots, following the guidelines suggested by Higgins et al. (2003). Each included study was independently assessed, and the results were tabulated in a risk-of-bias matrix. This process was conducted by a single reviewer, so no disagreements required resolution with other researchers. However, due to the small number of studies, formal statistical tests such as Egger's regression were not performed.

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) system was used in this study to assess the quality or certainty of the evidence obtained from the included systematic reviews and meta-analyses. This standardized tool allowed us to assess the reliability and robustness of the conclusions drawn from the analyzed studies, facilitating an informed interpretation of the results. GRADE classifies the certainty of the evidence into four main levels: high, moderate, low, and very low. Categorizations are made according to the degree of confidence that the results will remain stable in the face of future research. This ensured a rigorous assessment of the robustness of the evidence, contributing to better decision making based on reliable data. Thus, for the methodological assessment, the criteria comprising the checklist had a general or specific focus, in the latter case, adapted to the quantitative approach used (Eadie et al., 2018). Open-ended questions were operationally defined to provide a yes or no response, using the following 8 items: Item 1—did the study address a clearly focused topic? Item 2—was the cohort recruited in an acceptably accurate manner? Item 3—was the outcome measured accurately to minimize bias? Item 4—did the authors identify all important confounders? Item 5—did the authors account for confounders in the design and/or analysis? Item 6—was the follow-up of participants sufficiently complete? Item 7—was the follow-up of participants sufficiently long? Item 8—were the results accurate (e.g., do they report confidence intervals, standard errors, or standard deviations)?

The risk of bias assessment was performed by a single reviewer, with no disagreements to resolve. Due to the small number of included studies, formal statistical tests for publication bias such as Egger's regression were not performed.

3. Results

Firstly, the search formula and the inclusion and exclusion criteria resulted in a total of three studies for the systematic and meta-analytic review.

3.1. Descriptive Analysis of the Selected Studies

Firstly, regarding the sample sizes, they ranged from $N = 23$ participants (Lai et al., 2019) to $N = 30$ (C. H. Chen et al., 2014; M. H. M. Chen et al., 2019). Altogether, a total of 84 students participated in the experimental groups and 82 participated in the control groups. Regarding age, the studies by M. H. M. Chen et al. (2019) and Lai et al. (2019) indicated that the participants were between 10 and 11 years old; meanwhile, in the study by C. H. Chen et al. (2014), the participants were aged 12 years. All the studies included both boys and girls. In terms of publication year, the oldest study was from 2014 (C. H. Chen et al., 2014), while the most recent ones were conducted just before the COVID-19 pandemic (M. H. M. Chen et al., 2019; Lai et al., 2019). Nevertheless, all the studies shared a common objective: to analyze the significance of ICT use in the learning of Natural Sciences in Primary Education students. Finally, regarding the publication location,

all the studies were conducted in Taiwan and published in English (C. H. Chen et al., 2014; M. H. M. Chen et al., 2019; Lai et al., 2019).

3.2. Synthesis of the Evidence Found

The reviewed studies show that the use of mobile technologies, adaptive learning systems, and personalized approaches based on learning styles can significantly improve students' academic performance. C. H. Chen et al. (2014) proposed a contextual learning approach based on progressive cues in the teaching of Natural Sciences in Primary Education, finding that this method not only improved learning outcomes compared to traditional one-stage systems but also encouraged greater effort from students in analyzing contextual information. Similarly, M. H. M. Chen et al. (2019) and Lai et al. (2019) developed an adaptive learning system that considers multiple dimensions of learning styles, such as the field dependency/independence model and Felder–Silverman dimensions. Their experiments demonstrated that those who learned through this adaptive system achieved better academic results than those who used conventional methods. Additionally, the development of a personalized educational game based on students' learning styles further confirmed that personalizing and adapting learning resources to individual student characteristics enhances their performance and engagement in the learning process.

3.3. Summary of Meta-Analytic Results

The meta-analytic findings on the use of ICT in Natural Science learning for Primary Education students, based on a total of $k = 3$ studies, selected 84 participants for the experimental group and 82 for the control group (Table 2). The results indicated low heterogeneity ($I^2 = 84\%$), with an effect size of $SMD = 0.47$, and a 95% confidence interval (CI) [0.06, 0.88]. Regarding significance, the results showed a significant effect ($p < 0.002$).

Table 2. Meta-analysis of the use of ICT in Natural Science learning.

<i>k</i>	<i>SMD</i>	95% Confidence Interval		Significance (<i>p</i>)	Heterogeneity	
		Lower Limit	Upper Limit		<i>Z</i>	<i>I</i> ² (%)
3	0.47	0.06	0.88	<0.002	2.26	84%

Below is the Forest plot of the effect size for the set of interventions for a better visual appreciation (Figure 3).

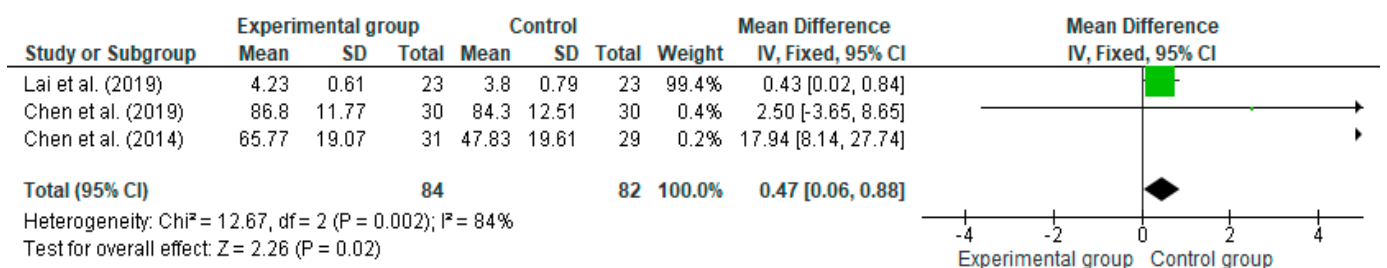


Figure 3. Forest plot of the effect size for the set of interventions. Note. τ^2 = variance estimate of effect sizes; χ^2 = existence of heterogeneity; df = overall indicator of effect size on the results of a meta-analysis; p = statistically significant heterogeneity; green dots = effect size of each study; black diamond on each variable = effect size of the set of studies; final black diamond = subgroup effect size; studies: C. H. Chen et al. (2014), M. H. M. Chen et al. (2019) and Lai et al. (2019).

3.4. Risk of Bias and Quality Criteria for the Included Studies

Once the studies for the meta-analysis were selected, the reliability of the results was assessed by evaluating the risk of bias, using the Cochrane risk of bias assessment tool. A

funnel plot (a graph used to verify the existence of publication bias) was also used. This allows us to observe the results of the risk of bias in the psychological parameters studied (Figure 4). The three quantitative studies reached 8/8 in the quality criteria for cohort study design, comprising factors such as recruitment of sub-objects, minimization of bias, control of confounding factors, and follow-up. In addition, to avoid a risk of bias in our assessment of the quality of the included studies, Items 9 and 10 were developed: Item 9—is there a relationship between the data and the conclusion? Item 10—what was the quality of the study design? Within Item 10, studies were determined to be of high, medium or low quality. The results can be seen in Table 3.

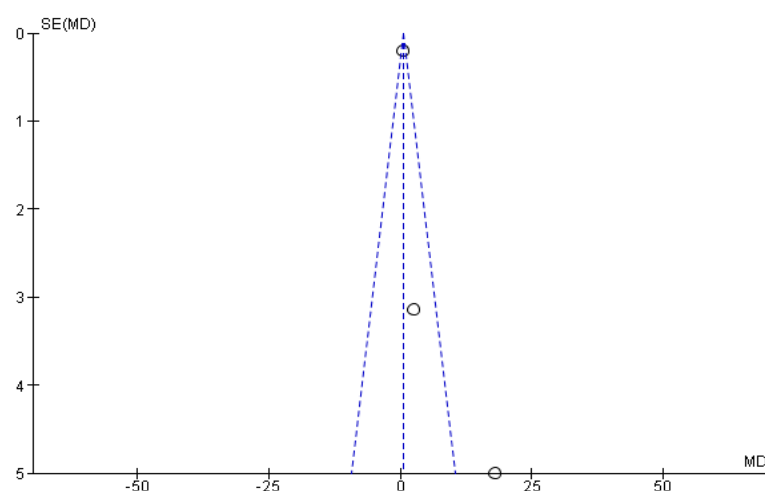


Figure 4. Overall risk of bias of the studies: C. H. Chen et al. (2014), M. H. M. Chen et al. (2019) and Lai et al. (2019).

Table 3. Risk of bias assessment of studies.

Author and Year of Publication	Item 1	Item 2	Item 3	Item 4	Item 5	Item 6	Item 7	Item 8	Item 9	Item 10
C. H. Chen et al. (2014)	✓	✓	✓	✓	✓	✓	✓	✓	✓	H
M. H. M. Chen et al. (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	H
Lai et al. (2019)	✓	✓	✓	✓	✓	✓	✓	✓	✓	H

Note. H = high; ✓ = explicitly present.

In this way, it was possible to determine that the study conducted by C. H. Chen et al. (2014) presented the highest risk of bias in each of the evaluated domains.

All studies showed a relationship between the data and the conclusion, demonstrating high quality in terms of study design.

3.5. Keyword Co-Occurrence Analysis

Regarding the co-occurrence of keywords from the conducted search, the analysis was carried out automatically using the VOSviewer software. Figure 5 presents a network visualization generated, which displays the co-occurrence of keywords extracted from the analyzed corpus. The nodes represent relevant terms or concepts, such as design, gender, game-based instruction, comprehension, among others. The lines connecting the nodes indicate co-occurrence relationships within the reviewed documents, allowing for the identification of thematic patterns. The nodes are grouped by color, highlighting two main clusters: the blue cluster includes terms such as gender, game-based instruction, e-textbook, and scenario simulation, mainly related to instructional design and educational

methods; and the green cluster, which encompasses terms such as comprehension, emotions, picture, text evidence, and display, linked to psychological aspects of learning and reading comprehension. The design node appears as a central element, acting as a connecting point between both clusters, emphasizing its fundamental role in bridging these thematic approaches.

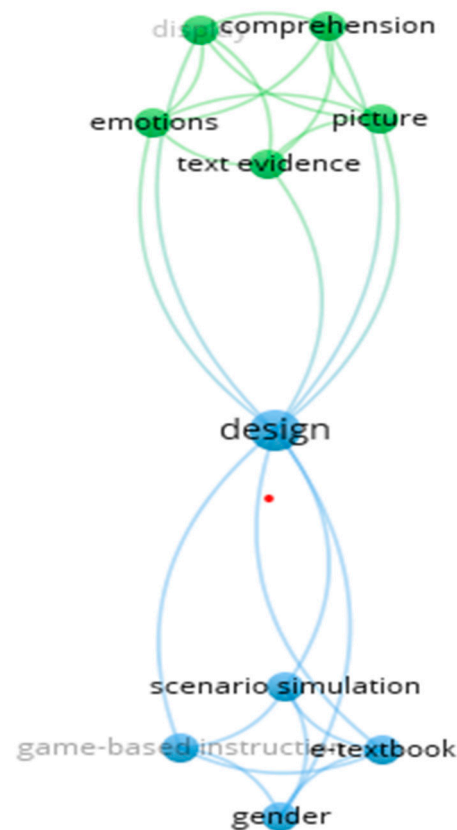


Figure 5. Thematic node map or clusters of the studies: [C. H. Chen et al. \(2014\)](#), [M. H. M. Chen et al. \(2019\)](#) and [Lai et al. \(2019\)](#).

The keyword co-occurrence analysis conducted with VOSviewer reveals two distinct thematic clusters—pedagogical design and psychological processes—highlighting ‘design’ as a central connecting element that bridges educational methods with the cognitive and emotional dimensions of inclusive science learning.

4. Discussion

The results obtained in this meta-analysis have successfully achieved the objective set out in this study, confirming the positive impact that ICTs have on learning Natural Sciences in Primary Education, especially in supporting students with a diversity of needs. The evidence gathered supports the idea that learning science at this educational stage promotes the development of critical thinking, scientific curiosity, and understanding of the immediate environment, becoming a central pillar for fostering inclusive, equitable education adapted to the individual needs of each student ([Turan & Atila, 2021](#)).

The studies included in this meta-analysis agree that interventions in Natural Sciences are based on digital environments that combine interactive approaches (such as personalized educational games and augmented reality) and adaptive approaches (systems that adjust content presentation according to each student’s learning style). For example, [C. H. Chen et al. \(2014\)](#) implemented a system with progressive prompts that adapted to each student’s individual pace. Similarly, [M. H. M. Chen et al. \(2019\)](#) developed activities

with reasonable adjustments through the use of technology, enabling them to be adapted to all students. In this regard, to individualize and personalize teaching, they rely on the Felder–Silverman model, which is applicable in educational contexts related to science, technology, engineering, and mathematics (STEM) (Kohnke & Zaugg, 2025). These findings confirm the first research hypothesis, demonstrating that ICTs in Natural Sciences are primarily designed and applied as interactive and adaptive digital technologies aiming to address the diversity of needs that exist among students with SEN.

The meta-analytic results report a favorable effect for the experimental groups, showing that the use of ICT in Primary Education not only increases academic performance but also strengthens active participation and intrinsic motivation. Specifically, the contextual approach with progressive prompts encouraged greater analytical effort (C. H. Chen et al., 2014), and adaptive games increased emotional involvement and autonomy (Fernández-Batanero et al., 2018; Martínez, 2020), thus confirming the second research hypothesis.

The results support the idea that ICT use is effective in promoting inclusion and enhancing learning for students in Primary Education. Therefore, based on the previous literature, it appears that education assisted by augmented reality, educational software, or e-books on tablets or mobile devices generates greater motivation among students in relation to their learning. Use of ICTs leads to higher learning achievements compared to traditional methods. Other authors, such as Zhang et al. (2024), report that learning through ICT not only promotes digital skills but also supports creative self-efficacy among students with SENs. Moreover, practices using AI tools, like applications, robots, and simulations, can boost participation and support both social and academic progress. Personalized interventions for specific learning difficulties, such as dyslexia and dyscalculia, can be made through intelligent tutoring systems; research shows that positive learning outcomes can be obtained, supporting students who require different forms of support due to their specific educational support needs (Yang et al., 2024).

The use of ICTs in Primary Education for inclusive purposes in the learning of Natural Sciences is structured around two main thematic axes: on one hand, the design of an instructional pedagogical method and the development of technology-mediated strategies; and on the other, the cognitive and emotional processes involved in learning. The central position of the node linked to educational design indicates that it functions as a nexus between both dimensions, highlighting the importance of intentional and diversity-sensitive pedagogical planning as the foundation for effective inclusion through the use of ICTs in this area. In other words, this finding suggests that the effective use of ICTs to promote inclusion in Natural Sciences learning largely depends on a pedagogical method that incorporates instructional design by sequencing content and steps to facilitate structured learning. This design acts as a fundamental bridge between two key dimensions: first, the pedagogical strategies mediated by technology—i.e., how ICTs are integrated and employed in teaching processes—and second, the cognitive and emotional experiences students undergo during learning. Furthermore, the need for intentional and diversity-sensitive pedagogical planning is emphasized, deliberately addressing different individual needs and characteristics. Consequently, the implementation of this model could foster effective, meaningful, and high-quality educational inclusion aimed at achieving competency-based learning in Natural Sciences.

However, despite these results, this study is not without limitations. The main limitation may be a result of the databases used for document searching; we used WoS and Scopus, meaning that studies published on other databases may have been inadvertently omitted. Additionally, the limited number of articles included in the meta-analysis may have influenced the heterogeneity of the findings. Finally, regarding the methodological

quality of the included studies, it is important to note that the risk of bias was assessed through the visual inspection of funnel plots. However, no standardized tools such as RoB 2 or ROBINS-I were applied, which limited our ability to systematically evaluate the internal validity of the evidence. Additionally, as no formal tests for publication bias (e.g., Egger's test) were conducted due to the small number of studies, it is not possible to conclusively rule out the presence of such bias. These limitations should be considered when interpreting the findings, as potential biases in study design, reporting, or publication may influence the observed effect sizes.

This review outlines relevant practical implications of the research: we provide evidence of the positive impact of ICT on motivation, supporting diverse learning needs, and fostering learning. For this reason, it is suggested that educational institutions integrate these technologies into their curricula at the Primary Education stage through specific strategies such as interactive simulations, 3D content visualization, and immersive experiences that promote active learning. Furthermore, their use in collaborative and gamified activities can contribute to achieving educational objectives and energizing the teaching and learning process, benefiting both teachers and students. For future lines of research, it would be interesting for new systematic review and meta-analysis studies to analyze the potential impact of ICT in addressing diversity to promote inclusion, motivation, and personalized learning in other educational stages such as Early Childhood Education or Compulsory Secondary Education.

The learning of Natural Sciences in Primary Education, by promoting critical thinking, curiosity, and observational skills, finds a valuable ally in the use of ICT. These tools appear to enable the exploration of the environment through simulations, interactive resources, and virtual environments that expand opportunities for investigation and experimentation, while adapting to students' different learning styles and paces. In this sense, the use of ICT not only enhances scientific understanding but also seems to contribute to more effective attention to diversity by offering accessible and customizable resources that support the active participation of students with SENs, thus promoting a more inclusive and equitable education for all.

The use of ICT in education offers multiple advantages, such as facilitating access to diverse and up-to-date educational resources, promoting active and interactive methodologies that enhance motivation and meaningful learning, and facilitating the ability to adapt teaching to different learning paces and styles. Additionally, ICT enables connection and collaboration between students and teachers beyond the physical space, expanding educational opportunities. However, it also presents disadvantages, including the digital divide that limits equitable access to these technologies, dependence on technical resources and connectivity, and the risk of distraction or overexposure to non-educational content. Furthermore, the effective implementation of ICT requires ongoing teacher training and appropriate pedagogical design to avoid superficial or inefficient use (Soriano-Sánchez & Jiménez-Vázquez, 2025).

Finally, the results of this study suggest the significance of ICT as a catalyst for active, inclusive, and student-centered methodologies that promote the achievement of meaningful learning in Natural Sciences. Based on the analyzed evidence, it can be reaffirmed that there is a need to continue promoting educational policies that encourage teacher training in digital and pedagogical competencies, ensuring effective and sustainable implementation of technologies in classrooms; this is especially relevant in contexts where student diversity demands flexible, innovative, and equitable educational responses that allow for the personalization and individualization of teaching to achieve competency-based learning. Inclusive excellence is an ongoing journey, not a destination, and it requires

leadership commitment as well as faculty, staff, and student involvement to create a diverse environment where all individuals feel valued and respected.

5. Conclusions

This meta-analysis highlights the importance and positive impact of ICTs on the learning of Natural Sciences in Primary Education, especially in addressing student diversity. The reviewed evidence confirms that personalizing learning through adaptive systems, using methodologies based on learning styles, and employing innovative technologies significantly enhance academic performance, motivation, and student engagement. Furthermore, it is evident that learning sciences not only enriches scientific knowledge but also contributes essential elements to the development of critical thinking, curiosity, and understanding of the environment—fundamental pillars for inclusive, equitable, and quality education.

The effective use of ICTs in Primary Education for inclusion in the learning of Natural Sciences depends on instructional design as an essential link between technological strategies and cognitive-emotional processes, highlighting the importance of intentional pedagogical planning adapted to student diversity.

In sum, the effective integration of ICTs in Primary School classrooms is, therefore, presented as a necessary pathway for addressing diversity, ensuring the right of all students to an education tailored to their individual needs which enhances their abilities. Ultimately, it is essential to continue promoting teacher training in the pedagogical use of technologies and to design innovative educational strategies that make the most of ICT's potential to transform classrooms into truly inclusive, equitable, motivating spaces for all students, thereby transforming the educational reality in Primary Education to guarantee high-quality education that responds to the diversity of interests, abilities and motivations that exists among the student population.

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Informed Consent Statement: This meta-analysis was conducted following the ethical guidelines established for secondary research. Since the data used comes from previously published studies, it was not necessary to obtain direct informed consent from the participants. However, fundamental ethical principles such as transparency, integrity, and respect for the original authorship of the included studies were ensured. To minimize bias and ensure the validity of the results, only studies that met rigorous methodological standards and had been approved by the relevant ethics committees in their respective contexts were selected. Additionally, the analysis and presentation of the data were conducted objectively, without manipulation or omission of relevant information. Furthermore, the principles of the Declaration of Helsinki and guidelines set by organizations such as the Committee on Publication Ethics (COPE) were followed for the review and synthesis of scientific literature. Any misuse of data was avoided, and each source was properly cited to preserve the academic integrity of the study.

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